KILIAN MINKWITZ

AI Engineer · ML Engineer · RAG, GraphRAG, Search & Cloud-Native Data Platforms

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Right to work: UK Student visa · Graduate Route after MSc · no sponsorship required until 2029

PROFILE

MSc Artificial Intelligence at Queen Mary University of London (currently ~77% average), with paid data engineering experience at Lebara and strong fit for early-career AI Engineer, ML Engineer, Data Scientist and Data Engineer roles in London. Hands-on focus across LLM/RAG, Neo4j-backed GraphRAG, GNNs, information retrieval, NLP, applied ML and production Spark/Kubernetes data platforms; BSc Data Science specialisation with module 1.0 grades in mathematics for data science, numerical methods, nonlinear optimisation and software engineering.

KEY ACHIEVEMENTS

- **Spark-on-AKS Datalake 2.0 —** Built a self-hosted Spark proof of concept on Azure Kubernetes Service at Lebara that became the basis for the company's Datalake 2.0 migration; travel-eSIM data already runs through the new architecture in production, which saves Lebara hundreds of thousands of pounds per year.
- **Applied ML uplift —** Improved 30-day app-install prediction AUC from 0.63 (logistic-regression baseline) to 0.78 with a PySpark Gradient-Boosted Tree classifier on real web-traffic data.
- **GraphRAG research —** Building an LLM-backed Neo4j GraphRAG / G-Retriever pipeline benchmarking rule-based, classical-ML, MLP, GCN, GAT and SAGE/R-GCN retrievers with Hit@k, AUROC, Answer F1, fidelity and GNN-Explainer-driven path-disruption metrics — testing when message passing actually helps retrieval.

EXPERIENCE

Junior Data Engineer (Internship) — Lebara Mobile Group, London, UK — Feb 2025 – Sep 2025

Full-time internship, extended through Bachelor thesis work · Data & Insight (Data Engineering and Data Science)

- Built a proof-of-concept platform to reduce Databricks dependency by running selected Spark transformations on Azure Kubernetes Service — custom Docker images, ADLS Gen2 access, Delta table reads/writes, and Spark master/worker deployment.
- Developed a data-science sprint predicting whether web-only customers install the app within 30 days; improved AUC from 0.63 (logistic-regression baseline) to 0.78 with a PySpark Gradient-Boosted Tree classifier.
- Implemented PySpark ETL pipelines from Bronze to Silver layer, adding foreign keys, derived values and Delta outputs for real Lebara web-traffic and transaction workflows.
- Designed Airflow orchestration for spinning Spark clusters up, running jobs and tearing them down, with CI/CD and infrastructure workflows supported through Azure DevOps.
- Proof of concept is now being expanded by Lebara into its Datalake 2.0 platform; further platforms are planned for migration onto the new architecture.
- Shared ownership via an incident playbook for daily pipelines and a knowledge-transfer session, “Spark on AKS in 45 Minutes,” delivered to the wider data team.

EDUCATION

MSc Artificial Intelligence — Queen Mary University of London — Sep 2025 – Sep 2026

- Current average ~77% · Dissertation: *GNNs in GraphRAG Pipelines: When Do They Help and Can We Verify How?*
- Selected modules: Machine Learning, Reinforcement Learning, Neural Networks and NLP, Information Retrieval, Conversational Agents and Dialogue Systems, AI Ethics.

BSc Computer Science — Hochschule Bonn-Rhein-Sieg, Germany — Sep 2022 – Aug 2025

- Final grade 1.7 (German scale: 1.0 = best, 4.0 = pass) · 180 ECTS · Specialisation: Data Science.
- Bachelor thesis: *Assessing the Feasibility of Predicting Tariff Performance in an MVNO Enterprise Product Catalogue* — thesis 1.3 (very good) · colloquium 1.0 (very good).
- Selected module grades: Mathematical Algorithmic Foundations for Data Science (1.0), Numerical Methods for Simulation (1.0), Nonlinear Optimization (1.0), Software Engineering I (1.0), Databases (1.3).

Selected AI & Engineering Projects

- **MSc Dissertation — GNNs in GraphRAG Pipelines** / *GitHub* — Controlled LLM-backed GraphRAG / G-Retriever-style pipeline on Neo4j infrastructure comparing rule-based, logistic-regression, tree, MLP, GCN, GAT and SAGE/R-GCN retrievers while holding graph, splits, features, LLM and evaluation constant. Evaluating retrieval and answer quality with Hit@k, AUROC, Answer F1, fidelity and GNN-Explainer-based path-disruption metrics to test when message passing actually helps retrieval.
- **Multi-Domain University Retrieval Engine** / *GitHub* — QMUL Information Retrieval system across six academic and professional domains using domain BERT encoders, sentence embeddings, FAISS (with a Chroma prototype), BERT query routing, dense + BM25 retrieval, RRF fusion and optional ColBERT reranking; LLM components served through Hugging Face and Ollama.
- **Bachelor Thesis — Telecom Tariff Performance Prediction** *(Grade 1.3, very good)* — Spark-native ML workflow predicting 30-day prepaid tariff uptake from EPC product metadata, promotions, CRM activity, OCS transactions and external market signals. A Gradient-Boosted Tree achieved 40.22% MAPE vs. ~48% naive baseline, with external variables adding a further 2–3 percentage-point improvement.
- **Statistical Planning & Reinforcement Learning Coursework** / *GitHub* — Implemented and evaluated planning / RL experiments in Python and Jupyter as part of the QMUL MSc AI curriculum.
- **Feed-Forward Neural Network from Scratch** / *GitHub* — Object-oriented feed-forward neural network with backpropagation — each neuron as its own instance — experimenting with Adagrad, RMSprop and Adam optimisers.
- **MVNO Copilot — Agentic GraphRAG Assistant (learning project)** / *GitHub* — In-progress project building an agentic telecom knowledge assistant: a LangGraph agent planning hybrid retrieval over a Neo4j graph and Qdrant vector store, served via FastAPI, with a LoRA-fine-tuned query router, Langfuse + ragas evaluation and guardrails, targeting AWS Bedrock for generation. Foundation in place — typed, mypy-strict, tested and CI-gated Python; agent, serving and evaluation components in active development.

Technical Skills

AI / ML — PyTorch, TensorFlow, scikit-learn, pandas, NumPy, XGBoost / GBT, Transformers, Hugging Face, Generative AI, GNNs, NLP, Reinforcement Learning, model evaluation.

LLMs / RAG / Search — LLMs, RAG / GraphRAG, Neo4j, Ollama (local serving), Hugging Face Hub, OpenAI API, G-Retriever, GNN-Explainer, embeddings, vector search (FAISS, Chroma), BERT encoders, BM25, ColBERT, RRF, Hit@k, AUROC, Answer F1.

Data Engineering — Spark, PySpark, Delta Lake, Databricks, Azure Data Factory, NiFi, Airflow, PostgreSQL, MySQL, ETL / ELT, data lakes, feature engineering.

Cloud / DevOps — Microsoft Azure, AKS, Docker, Kubernetes, Azure DevOps, Git, GitHub, Linux, container registries, CI/CD.

Programming & BI — Python, Java, SQL, C++, C#, C, JavaScript, x86 Assembly · Power BI, Azure Synapse.

Currently developing — LangChain / LangGraph agents, FastAPI model serving, LLM evaluation & observability (Langfuse, ragas), LoRA / QLoRA fine-tuning (Hugging Face PEFT), LLM inference optimisation.

Additional

Languages — German C2 (native-level) · Spanish C2 (native-level) · English C2 · French B2.

Interests — Judo since age 5, West-German runner-up · skiing · travel (~45 countries) · wine and sommelier basics.

References — Available on request, including former Lebara manager and external Bachelor thesis supervisor.